

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Mechanical Engineering Semester : V Subject Code & Name: BTMEC501 & Heat Transfer Max Marks: 60 Date: Duration: 3 Hr.			
	Instructions to the Students: 1. All the questions are compulsory. 2. Use of Heat Transfer data book is allowed. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.			
A)	A gas filled tube has 2 mm inside diameter and 25 cm length. The gas is heated by an electrical wire of diameter 50 microns(0.05 mm) located along the axis of the tube. Current and voltage drop across the heating element are 0.5 amps and 4 volts, respectively. If the measured wire and inside tube wall temperatures are 175 °C and 150 °C respectively, find the thermal conductivity of the gas filling the tube.	L3/CO1	06	
B)	Derive an equation for heat conduction through composite cylinder with neat sketch.	L1/CO1	06	
C)	An electric hot plate is maintained at a temperature of 350 °C, and is used to keep a solution boiling at 95 °C. The solution is contained in a cast-iron vessel of wall thickness 25 mm, which is enamelled inside to a thickness of 0.8 mm. The heat transfer coefficient for the boiling solution is 5.5 kW/m ² K, and the thermal conductivities of the cast iron and enamel are 50 and 1.05 W/mK, respectively. Calculate: (i) The overall heat transfer coefficient. (ii) The rate of heat transfer per unit area.	L1/CO1	06	
Q.2	Solve Any Two of the following.			
A)	Write the assumptions made for the analysis of heat flow through fin. Derive an equation to find heat dissipation from a fin insulated at the tip.	L1/CO3	06	
B)	Derive an equation to find critical thickness of insulation for sphere.	L1/CO2	06	
C)	What is meant by transient heat conduction? Explain the significance of Heisler charts in solving transient conduction problems.	L2/CO2	06	
Q. 3	Solve Any Two of the following.			

A)	What are the assumptions for momentum equation for hydrodynamic boundary layer over a flat plate, derive an expression for the same.	L1/CO4	06
B)	Air at 20 °C is flowing over a flat plate which is 200 mm wide and 500 mm long. The plate is maintained at 100 °C. Find the heat loss per hour from the plate if the air is flowing parallel to 500 mm side with 2 m/s velocity. What will be the effect on heat transfer if the flow is parallel to 200 mm side. The properties of air at $(100+20)/2=60$ °C are $k= 0.025$ W/m °C and $Pr=0.7$, $\nu=18.97 \times 10^{-6}$ m ² /s.	L3/CO4	06
C)	Calculate the heat transfer from a 60 W incandescent bulb at 115 °C to ambient air at 25 °C. Assume the bulb as sphere of 50 mm diameter. Also, find the percentage of power lost by free convection. The correlation is given by: $Nu= 0.60 (Gr \cdot Pr)^{1/4}$	L3/CO4	06
Q.4	Solve Any Two of the following.		
A)	Explain briefly the various regimes of saturated pool boiling with neat sketch.	L1/CO5	06
B)	Derive an equation to determine LMTD for parallel flow heat exchanger.	L1/CO5	06
C)	An oil cooler for a lubrication system has to cool 1000 kg/h of oil ($C_p=2.09$ kJ/kg °C from 80°C to 40°C by using a cooling water flow of 1000 kg/h at 30°C. Give your choice for parallel flow or counter flow heat exchanger, with reasons. Calculate the surface area of the heat exchanger, if the overall heat transfer coefficient is 24 W/m ² °C. Take C_p of water =4.18 kJ/kg °C.	L3/CO5	06
Q. 5	Solve Any Two of the following.		
A)	For a hemispherical furnace, the flat floor is at 700K and has emissivity of 0.5. The hemispherical roof is at 1000K and has emissivity of 0.25. Find the net radiative heat transfer from roof to floor.	L3/CO6	06
B)	State and prove reciprocity theorem with neat sketch.	L1/CO6	06
C)	Two large parallel plates with $\epsilon =0.5$ each, are maintained at different temperatures and are exchanging heat only by radiation. Two equally large radiation shields with surface emissivity 0.05 are introduced in parallel to the plates. Find the percentage reduction in net radiative heat transfer.	L3/CO6	06
	*** End ***		